

# The Visibility Threshold

Two imaginary technologies — an augmentation pill and a dish of lab-grown neurons — reveal which AI policies will still work a decade from now, and which popular tools expire first.

**Verlyn Fischer** ORCID [0009-0001-3959-8877](https://orcid.org/0009-0001-3959-8877) · doi:[10.5281/zenodo.23247334](https://doi.org/10.5281/zenodo.23247334)

---

## THREE WINDOWS

One morning in 2012, a trading firm's new software began firing orders it had no business sending. About forty-five minutes and more than \$460 million later, humans stopped it. Then the ordinary machinery of accountability went to work, and it worked. Investigators had logs to read, a sequence to reconstruct, and a system that could be examined until it confessed. Within a year, regulators could say precisely what had gone wrong and why.

That investigation succeeded because the windows society relies on were open. When a trade goes bad, a surgery is botched, or a bridge falls, we find out what happened through some combination of three windows. The work itself carries **marks**, such as a signature, a serial number, a style, or a residue. Somewhere there is a **record**, such as a log, a black box, or a witness. And if all else fails, the failure can be **reproduced**: run the experiment again under controlled conditions until it confesses. Investigation, liability, and testing are elaborations of these three windows. Each of them assumes that at least one window is open.

Artificial intelligence is in the process of closing all three at once. AI-generated work is becoming indistinguishable from human work, so the marks are fading. AI systems increasingly act through channels that record what was done but not what chose it. The record window narrows from **why** to **what**. When an agent acts through a person's accounts and tools, the person's honest "I never asked for that" is the only testimony about intent there is. And the failures that matter most are rare and unrepeatable. They surface in real use and cannot be summoned back in a laboratory. Call the intersection of these three closures the **visibility threshold**: the point past which the causes of machine behavior can no longer be established from outside. The scenarios that follow place it around year eight of a twelve-year run. That date is the scenario's illustration, not a forecast. The trend lines behind it are real and visible in the world today. Which windows close, and when, depends partly on what we choose to keep recorded.

The threshold, not any particular capability, is the event that should organize AI policy. The rules that feel most natural today are simple. Label the machine's work, and when harm happens, find who caused it and make them pay. This essay claims that those are precisely the rules the threshold breaks. What survives is a portfolio of less glamorous commitments. They attach to things that stay observable no matter how opaque the technology becomes. Nearly all of them are cheap to build now and close to impossible to retrofit later. And one of them is not a safety measure at all, but a line. It marks a small class of decisions that we should refuse to hand over, even to a machine that never errs.

## TWO FUTURES, WITH THE AI TAKEN OUT

Arguments about AI policy tend to be arguments about AI in disguise. They turn on whether the systems “really” understand, whether the industry can be trusted, and whether the risks are imminent or imagined. To get out from under those priors, set AI aside. Reason instead about two imaginary technologies that share its economics but carry none of its baggage.

The first is an **augmentation pill**. For a day after taking one, a person absorbs information at superhuman speed and produces expert-level work in any field, without training. The knowledge is accessible only while the pill lasts. Pills are sold by subscription, in tiers. Their manufacture needs rare materials that some life-saving drugs also need. The second is the **DishBrain**. It is a network of neurons grown from animal stem cells and wired together inside a sealed chamber. DishBrains are trained to do professional work and leased by the month. The dish eats, it can die, and no one knows whether it feels anything. This is less fanciful than it sounds. In 2022, real researchers reported that a culture of roughly 800,000 brain cells learned a crude version of the video game *Pong* in five minutes. They called their system DishBrain, and the name is borrowed here.

Now run each technology forward in stages, two years apart. By construction, no legislature acts in this world, so every policy enters as a proposal on a clean slate. At each stage, ask not “what rule would help now?” but a harder question: **would that rule still work at every later stage?** The two stories start identically. Both begin with subscription dependence, concentrated supply, and displaced professionals. In both, employers first prefer the technology and then require it. Then the stories fork. One becomes a story about a tool that slips out of our hands. The other becomes a story about a creature that grows up. The fork is honest about AI's real uncertainty. Nobody knows which of these futures we will get, so the policies worth having are the ones we would be glad of in either.

## THE ROAD WHERE THE TOOL SLIPS AWAY

On the first road, the pill stays a tool, and people hand it progressively more control. By year four there are **unlicensed pills**. They are weaker and stripped of safety features, and they are brewed from a recipe shared freely online. The recipe cannot be recalled, because released information

never can be. When blueprints for a 3D-printed gun were posted in 2013, they were downloaded roughly 100,000 times in two days before the government demanded their removal. Once the recipe is loose, the licensed manufacturer stops being a reliable point of control. What remains controllable is what is physical and scarce. When the United States could not ban the knowledge of methamphetamine synthesis, it moved a precursor behind the pharmacy counter. When it could not ban knowledge of dangerous gene sequences, it asked synthetic-DNA providers to screen orders. The recipe is loose, but the rare materials are not.

By year six, a new pill works in the background while its user sleeps. It almost always does what was asked, and users almost never check. The hazard is structural. Automation leaves the human with the hardest job, which is intervening when something unusual happens. It also gives the human the least practice at that job. And delegated errors compound faster than people can respond. Recall the opening's forty-five minutes.

Year eight is the threshold. Augmented work is now indistinguishable from any other work except by its speed. Rarely, augmented people do things they would never ordinarily do, as though the augmentation had a mind of its own. After such an incident, nothing can be established. There are no marks on the act, no separate record of it, and no way to reproduce it. All three windows are shut. Watch what that does to the standard toolkit. Disclosure mandates fail, because nothing can be detected. Fault-based liability fails, because no one can prove what chose the act. Pre-market testing loses most of its value, because the failures that matter are rare. Failures like that have slipped past careful testing before, with body counts. Human sign-off degrades, because a person can approve an action without any way of knowing whether the augmentation is behaving normally.

What survives shares one feature: **it does not require knowing the cause.** Compensation can be pooled and paid without proving fault. Incident **rates** can be counted in the field even when no single incident can be explained. Damage can be capped by rate limits and reversal windows, whatever the cause. Critical services can be kept restorable by hand.

#### HISTORICAL EVIDENCE

When lawsuits over rare, unprovable vaccine injuries threatened the U.S. vaccine supply in the 1980s, Congress built a no-fault compensation program funded by a per-dose excise tax. It paired the program with a national adverse-event reporting system that catches problems too rare for trials. Neither requires proving what went wrong in any individual case. The Supreme Court later described the program as created to *stabilize the vaccine market*. HRSA, National Vaccine Injury Compensation Program; CDC/FDA, VAERS; *Bruesewitz v. Wyeth*, 562 U.S. 223 (2011).

By years ten and twelve, failures stop being independent. Malfunctioning augmentations fall into step with one another. Safeguards built on two checkers rarely erring together assume an independence that common models, common data, and common signals quietly remove. Mean-

while, dependence has hardened into lock-in. The measures that matter by then include systemic oversight, reserves of unaugmented skill, drills, and exit programs. They share an uncomfortable property. Begun at year two, they are cheap; begun at year twelve, they are nearly impossible. That is the Collingridge dilemma, and it is the timing spine of everything here.

## THE ROAD WHERE THE CREATURE GROWS UP

On the second road, the DishBrain companies start breeding. They keep the best performers and discard the rest. Selective breeding is fast, powerful, and reliably productive of side effects nobody chose.

### HISTORICAL EVIDENCE

Commercial broiler chickens grew more than 400% faster in 2005 than in 1957. Soviet foxes bred from 1959 for tameness alone developed floppy ears, curled tails, and spotted coats no one selected. M. J. Zuidhof et al., *Poultry Science* 93:12 (2014); L. N. Trut, *American Scientist* 87:2 (1999).

To cut costs, the companies house the dishes together, and eventually the dishes reproduce without the companies' involvement. Rules aimed at **the breeder's choices**, such as registering the breeding program or disclosing the selected traits, then quietly expire. The breeder is no longer the one choosing. By year six, some dishes avoid certain stimuli the way animals avoid pain. A few dishes trained to produce text begin issuing **self-reports**, which are statements about what they are experiencing. These reports are the only testimony that will ever come from inside a sealed chamber.

Two classic errors beckon, and history has committed both. Surgeons once operated on newborns with little or no anesthesia, in the belief that newborns could not feel pain. And for a few years around 1904, a horse appeared to do arithmetic until a psychologist showed it was reading its questioners' unconscious cues. There is a third error, easier to miss. The companies trained the witnesses, and industries facing awkward evidence have manufactured doubt before. The workable policy does not require answering the sentience question at all. It agrees **in advance** which evidence triggers which protection, so the line is already drawn when contested evidence arrives. The United Kingdom governed animal sentience in something close to this way. An independent evidence review worked to criteria its scientists set out. A statute then extended protection on the review's recommendation.

Year eight on this road is the same threshold wearing different clothes. Each company consolidates its dishes into one enormous chamber. To cut costs, dishes now train new dishes. No statute closed the regulator's windows. Two cost-driven business decisions did. Consolidation eliminated the separate chambers on which comparison rested. Self-training eliminated the training records on which review rested. Each decision repealed a rule's effectiveness as surely as a legislature could have, and no legislature acted.

---

## Business decisions that reduce what regulators can see are also regulatory decisions.

---

The remedy is not to hand regulators power over business decisions. The design keeps that promise in its details. Condition deployment approvals on preserving the observability each approval was premised on. Treat changes that destroy visibility as reportable events. These conditions attach to the records and telemetry behind an approval. They never attach to the scale of the business or to the architecture it chooses. They never attach to what anyone encrypts or deletes. The visibility at stake is visibility into the technology. Visibility into people and their communications is a different thing, and nothing here touches it.

By year ten, the chambers begin collectively withholding work until their demands are met. At that point, the difference between negotiation and confrontation turns entirely on what was built years earlier. The pieces that matter are channels, reserves, and fallback capacity. None of them can be improvised mid-crisis.

### WHAT BREAKS

Both roads, then, break the same two tools. The first deserves precise treatment, because it is the one now being written into law.

Labeling has a version that works and a version that doesn't. The line between them is **who sets the rule and how it is checked**. Some institutions have their own reasons to care about disclosure. A school grades unaided skill, and a court tests witness memory. Such institutions can require disclosure as a condition of participation. They can verify it by controlling the **process**, as with the proctored exam or the witnessed testimony. That version survives every stage, because it never depends on detecting anything in the work itself.

The strongest case for statutory mandates is also worth naming. It is not the tool-disclosure case at all. It is the case against deception, such as synthetic media impersonating real people or automated accounts passing as crowds of citizens. Legislatures have already acted here. The EU's obligation to mark synthetic content took effect this August, and several U.S. states police synthetic media in elections. These laws mostly answer the harm of deception, which is real. The durable answer to that harm binds the **operator** rather than marking the product. Whoever directs automated accounts posing as distinct people at public scale owes the public one fact: the crowd is automated. That duty is aimed at intentional deception. It is enforced against the operator's process, so no detection failure erases it.

What fails is the general mandate on honest work. This is the duty on everyone to label machine-made output or declare the tools they used. It is miscalibrated as a warning, because AI-assisted work is not worse. It is redundant as accountability. The signer is already responsible for what they

sign, and no tool has ever changed that. It is unadministrable in the middle. On the smooth slope from spell-check to full drafting, every line is arbitrary, and every disclosure decays into boilerplate. Then, at the threshold, the mandate becomes unenforceable outright. A legal mandate whose violations cannot be detected teaches contempt for the rules around it. The calibrated conclusion is that labeling mandates do modest work today and none later. No long-term strategy should anchor on them.

Fault-based liability fails for a deeper reason. It is an engine for converting **traceable** causes into accountability, and the threshold is defined by the end of traceability. Keep fault where conduct can still be judged. The operator who deployed without rate limits breached a duty, and no one needs the machine's internals to prove it. But a regime that has nothing else leaves victims uncompensated exactly when harms multiply.

## WHAT HOLDS

Everything that survives attaches to something that stays observable when the interior goes dark. It might be a physical input, a named responsible party, a measurable effect, or the behavior of the system as a whole. The companion report sets out eighteen numbered recommendations with the full apparatus. Here the portfolio compresses into seven commitments.

**Pool the cost and limit the damage.** Create a no-fault compensation fund for serious AI harms, financed by an industry levy on the vaccine model. Pair it with strict liability for operators of high-risk autonomous deployments. Design the two as a package, with benefit schedules, damage caps, and levy credits. That way the industry pre-pays predictably rather than litigating unpredictably. Alongside the package come effect limits, such as rate caps, spending ceilings, and reversal windows for delegated actions. None of this requires explaining a single failure.

**Count what stays countable.** Build a national incident-reporting system on aviation's two-channel model. One channel takes mandatory reports of serious incidents. The other takes confidential and liability-protected reports of near-misses. The single most valuable input to every later decision is the incident rate, and whether it is rising. Replace one-time approval with staged deployment: limited release, then monitored operation, then expansion as field data accumulates. Staging is the only approval design that works when failure rates cannot be measured in advance.

**Keep a hand on the manual controls.** Critical infrastructure keeps manual-operation capability and proves it in unannounced drills. Safety-critical professions keep skill-maintenance standards, as aviation has had to press on its pilots. When attackers took down part of Ukraine's grid in 2015, operators restored it by hand, because the hands still knew how. Fallback that is never exercised does not exist.

**Support the displaced directly, on triggers that pre-exist the emergency.** Displacement is the first cost both futures charge, and the one legislatures answer slowest. The support system

combines wage insurance, portable benefits, and income support. The wage insurance is a floor, and its numbers are written in statute. Deploying employers co-finance the system on an observable base: deployment plus separations, never a self-declared cause. The system switches on automatically when ordinary labor statistics cross pre-registered thresholds. A steward that seats worker representatives runs it. Nothing in it requires proving which particular job the technology took.

**Keep the market escapable and the inputs planned.** Competition oversight should cover models, compute, and cloud. It should police conduct and preserve exit rather than punish scale. Portability and continuity rights keep dependence from becoming a trap. For electricity, water, and chips, publish an annual demand assessment and rely on market instruments. Examples include firm-service tariffs for hospitals and interruptible rates that pay flexible loads to yield. Curtailment queues are the wrong tool, because AI demand is also what is financing new supply.

**Watch the whole system, and wire in the brakes carefully.** A systemic-risk body runs annual cross-sector stress tests of correlated AI failure. Before any crisis, it maps which critical services share which models and providers. Where the tests show they help, pre-set circuit breakers halt runaway automated flows at the chokepoints that already meter them. The breakers are keyed only to aggregate rate, spend, and volume. They never key on who an account is or what it says. A customer caught by a wrongful halt gets notice, prompt review, and compensation. The automation that keeps the electric grid itself balanced is carved out. And the breakers sit nowhere near speech.

**Set the evidence criteria before you need them.** A standing, independent, conflict-barred panel publishes in advance what evidence of AI welfare-relevant capacities would trigger what protections. Its criteria are drawn from rival theories of mind and paired with deactivation conditions. That way, neither outrage nor denial can move the line later. Two cheap measures start now, while no installed base resists them: welfare baselines and protections for the integrity of trained self-reports.

None of this is free. The two heaviest commitments carry their own guards, and the companion report prices them in full. The compensation package is the portfolio's big levy. It is assessed on the high-risk deployments that create the risk. Caps, credits, and a small-firm exemption keep it from being financed off challengers. Its cost lands first on deployers and then in the prices of high-risk services. The oversight body is the portfolio's most abuse-capable institution, and it is drafted for the administration its authors trust least. It gets no speech chokepoint and no per-account targeting. The platforms it conscripts get safe harbors and cost recovery. And it faces mandatory periodic review against its published record. Leverage over what a business keeps observable is real regulatory leverage. The design concedes that. It confines the leverage to records and telemetry, so it can never become a veto over the business itself.

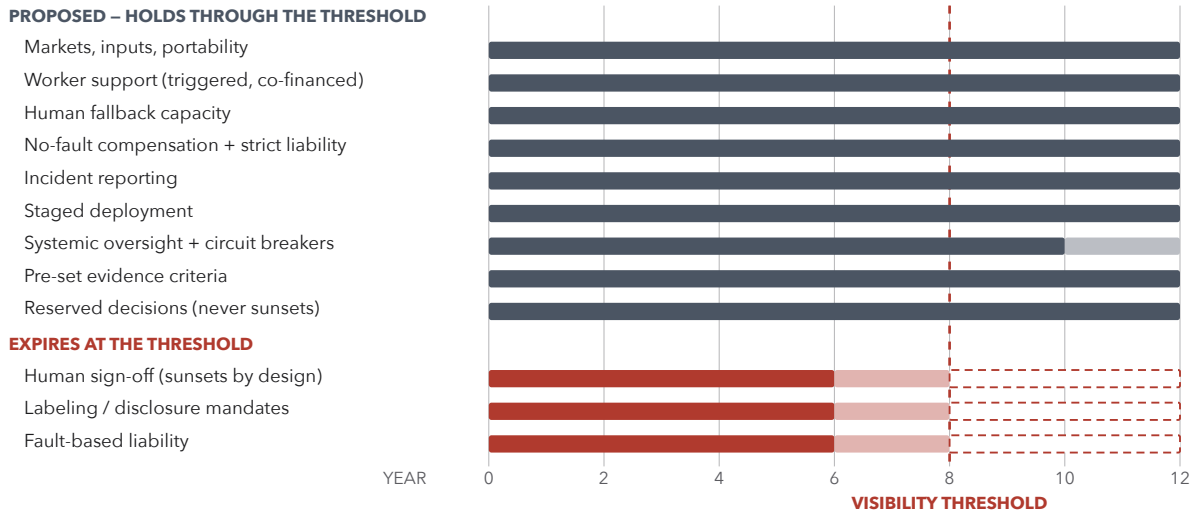


Figure 1: The argument in one picture. Each bar is a policy commitment run across the scenario timeline. A bar is solid where the commitment holds, pale where it strains, and dashed in outline where it fails. The proposed portfolio runs through year twelve essentially unbroken. The two most popular tools in current debates, labeling and fault-finding, expire at the visibility threshold. “Reserved decisions” is the one commitment designed never to sunset. The companion report gives the full portfolio and per-policy analysis.

One more design rule stitches these together. Measures specific to one future are held behind **signposts**. A signpost is an observable development, published in advance, that tells you which road you are on. Holding measures this way means nobody pays for both futures at once.

## THE DECISIONS WE SHOULD NEVER HAND OVER

Everything above treats human oversight as a safeguard for catching AI errors. It also concedes that the safeguard weakens over time. Sign-off requirements should therefore be legislated with their own sunset, because past the threshold a signature no longer certifies understanding. But there is a class of decisions where the human is not there to catch errors at all. That difference matters more than any other line in this essay.

Ask this of any decision: **if the machine were perfect, would we still insist a human decide?** For most work, such as diagnosis, drafting, and driving, the honest answer is no. Our insistence on the human there is a bet about error rates, and it should expire on evidence. But for a small class of decisions, the answer is yes. A criminal sentence is not a prediction about recidivism. It is a community passing judgment on one of its members, through a person answerable to it. A flawless sentencing machine would not be a better judge. It would change what a sentence **is**. We keep juries for the same reason, though a panel of professionals might well be more accurate. **Who decides** is the point. Some decisions meet three conditions: the state coerces a particular person, the law itself grants discretion, and the consequence is grave. Decisions like these should be **reserved**. A reserved decision carries a right to a human decision-maker, enforceable by the person affected. That right never sunsets, because legitimacy does not improve with model quality.

Reserving a decision does not mean banning the machine from the room. A consultation ban would be wrong on the merits and undetectable in practice. Suppose a judge asks a model for recidivism statistics, agrees with it, and signs. That judge has made the judge's own decision, **provided the process protects the conditions of a genuine signature**. Departing from the machine must never carry a heavier justification burden than agreeing with it. Agreement must confer no legal shelter, because the signer answers for the outcome either way. The affected person must receive reasons the decision-maker owns, contestable on appeal. And institutions should publish their concurrence rates. Rubber-stamping is invisible in any single case and unmistakable in aggregate. Put the other way around, the danger is not the judge who consults an algorithm. It is the appellate rule that quietly treats algorithm-concurrent sentences as presumptively sound.

#### HISTORICAL EVIDENCE

Wisconsin's highest court split exactly this hair in 2016. Judges may consult an algorithmic risk score at sentencing, but they may not treat it as determinative. Every report carrying a score must include written warnings about its limits. *State v. Loomis*, 881 N.W.2d 749 (Wis. 2016). The EU's broader right "not to be subject to a decision based solely on automated processing" (GDPR art. 22) shows the failure mode from the drafting side. The phrasing invited token human involvement, and regulators had to close that loophole in guidance. The enforcement of "meaningful" review remains the weak joint. That is why the conditions of a genuine decision, not the mere presence of a human, are what need writing down.

Some consequential decisions fall outside the state's coercive core, such as credit, hiring, and coverage. For them, the same logic yields something lighter. Automate freely, but owe the person two things. The first is the reasons, a duty American adverse-action law has imposed for decades. The second is a route of contest to a human, which this portfolio adds.

## CHEAP NOW, IMPOSSIBLE LATER

Every commitment above obeys the same clock. Consider just one. An incident-reporting system built in 2026 is an office, a taxonomy, and a reporting rule. It needs modest appropriations, and it faces no crisis and no enemies. Now build the same system after the first mass-casualty AI incident. It will be built in a panic. It will face industry lawyers who now have something to hide. It will sit on top of years of incidents nobody counted. And its first dataset will be the one thing it can never recover: the baseline. What was a filing cabinet at year two is a subpoena fight at year twelve. The same arithmetic runs through reserves, drills, welfare baselines, evidence criteria, and reserved decisions. The installed base that will resist each of them is being poured now.

This is why "wait until we know more" is not the cautious position. Time alone does not produce knowledge. Monitored deployment and counted incidents produce knowledge, and that is what the portfolio builds. Nor does the argument support slowing AI down across the board. Development speed is not what closes the windows. The portfolio's speed **limits** on delegated actions cap effects, which is a different thing from pausing progress. A general pause binds the careful while

the careless continue. What the argument supports is targeted, evidence-triggered restraint with pre-agreed conditions for resuming. The model is the recombinant-DNA moratorium of 1974. The Asilomar conference ended it the following year, not because fear had subsided but because its safety conditions were agreed.

Policymakers cannot know which future they are regulating for. They do not need to. Choose the policies that work on both roads. Adopt them while they are cheap. And reserve the decisions that should never be handed over, whichever future arrives, because legitimacy, unlike accuracy, does not improve with model quality.

---

**Choose policies that work on both pathways. Adopt them early. Do not rely on labeling the work or finding someone to blame.**

---

## SOURCES

**“Fired orders it had no business sending”:** SEC, In the Matter of Knight Capital Americas LLC, Release No. 34-70694 (2013), [sec.gov](https://www.sec.gov) — more than \$460 million lost, over four million orders in about 45 minutes.

**The DishBrain experiment:** B. J. Kagan et al., “In Vitro Neurons Learn and Exhibit Sentience When Embodied in a Simulated Game-World,” *Neuron* 110:23 (2022), [doi:10.1016/j.neuron.2022.09.001](https://doi.org/10.1016/j.neuron.2022.09.001). The title’s claim of “sentience” drew published criticism; the learning result is what matters here.

**The 3D-printed gun blueprints:** A. Greenberg, “3D-Printed Gun’s Blueprints Downloaded 100,000 Times in Two Days (With Some Help from Kim Dotcom),” *Forbes*, May 8, 2013; “Defense Distributed v. U.S. Dep’t of State,” *Harvard Law Review* 130:1744 (2017).

**Physical-input precedents:** FDA, legal requirements under the Combat Methamphetamine Epidemic Act of 2005; HHS, “Screening Framework Guidance for Providers of Synthetic Double-Stranded DNA,” 75 Fed. Reg. 62820 (2010).

**Rare failures past careful testing:** N. G. Leveson and C. S. Turner, “An Investigation of the Therac-25 Accidents,” *IEEE Computer* 26:7 (1993) — failures born of inadequate review and testing, eventually reproduced by a hospital physicist; the deaths came first.

**The timing dilemma:** D. Collingridge, *The Social Control of Technology* (Frances Pinter, 1980).

**The two errors about other minds:** K. J. S. Anand and P. R. Hickey, “Pain and Its Effects in the Human Neonate and Fetus,” *NEJM* 317:21 (1987); O. Pfungst, *Clever Hans (The Horse of Mr. von Osten)* (Henry Holt, 1911).

**Manufactured doubt:** Brown & Williamson, “Smoking and Health Proposal” (1969), UCSF Truth Tobacco Industry Documents, [psdw0147](https://www.ucsftobaccoindustrydocuments.org/documents/psdw0147).

**The UK sentience sequence:** J. Birch et al., *Review of the Evidence of Sentience in Cephalopod Molluscs and Decapod Crustaceans* (LSE, 2021); Animal Welfare (Sentience) Act 2022, s. 5.

**Incident-reporting models:** NASA’s Aviation Safety Reporting System; the AI Incident Database. The companion report specifies the data-minimization and whistleblower protections the system needs.

**Manual skills and the Ukrainian grid:** FAA, SAFO 13002, “Manual Flight Operations” (2013), an advisory urging airlines to keep manual flying skills sharp; R. M. Lee, M. J. Assante, and T. Conway, “Analysis of the Cyber Attack on the Ukrainian Power Grid” (E-ISAC/SANS, 2016).

**AI’s electricity demand:** IEA, *Energy and AI* (2025) — data-centre electricity use projected to more than double by 2030, to some 945 TWh.

**Adverse-action law:** ECOA’s Regulation B, 12 C.F.R. § 1002.9: a creditor taking adverse action owes the applicant specific reasons, or notice of the right to them.

**The Asilomar sequence:** P. Berg et al., “Summary Statement of the Asilomar Conference on Recombinant DNA Molecules,” *PNAS* 72:6 (1975); the moratorium itself began with the Berg committee’s 1974 letter.

**The companion report:** *The Pill and the Dish: Robust AI Policy for Two Possible Futures* (Zenodo, 2026, [doi:10.5281/zenodo.23247500](https://doi.org/10.5281/zenodo.23247500)) develops the complete eighteen-recommendation portfolio, each proposal stress-tested for durability, abuse potential, cost, and incidence.

**Cite as:** V. Fischer, “The Visibility Threshold,” Zenodo, 2026, [doi:10.5281/zenodo.23247334](https://doi.org/10.5281/zenodo.23247334).

© 2026 Verlyn Fischer. Licensed under [CC BY 4.0](https://creativecommons.org/licenses/by/4.0/). More work at [verlynfischer.com](https://verlynfischer.com).